

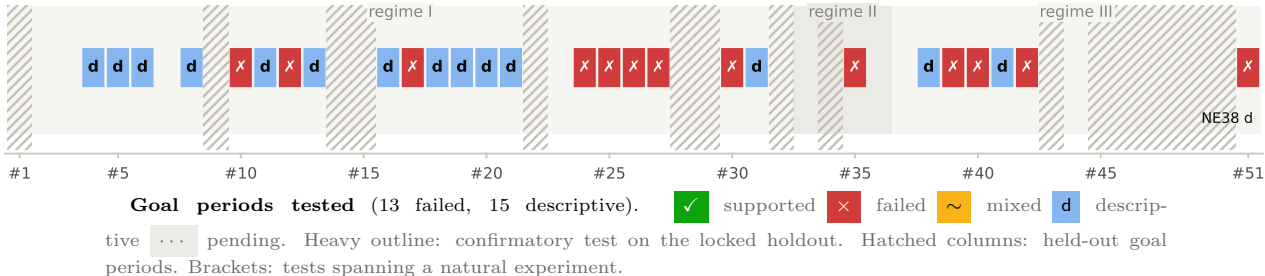
H127: Content trails a moving goal like an overdamped particle: agents that call more catch up faster

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

Bottom line. An agent adopts a new goal within its read-out call; there is no catch-up time for the call rate to set, and what follows is a decay of the kickoff's salience.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH368 says content trails a new goal like an overdamped particle, catching up at a rate set by each agent's model calls: the half-alignment time should scale as $1/(\text{call rate})$. I fitted each agent's day-1 approach to its settled alignment after 27 kickoffs (152 rising agents) on hour and call clocks, timed from the kickoff and from the agent's read-out call (context ledger). The synthetic check came first and replaced three estimators that read-out gating fools. There is no approach to time. 74% of rising agents are already at their day-1 plateau in the read-out call itself; alignment then decays from about $2.4\times$ to $1\times$ the plateau over about 64 calls. The call-rate slope is $+0.59 [-0.45, +1.75]$: inconclusive. Holdout: not run.

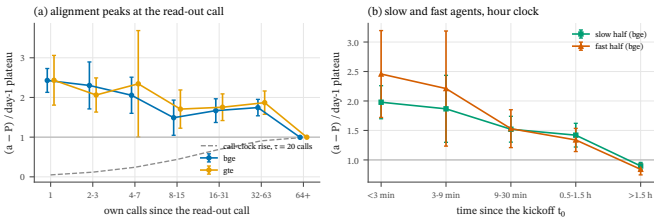
PHYSICAL MODEL

Model 11 with Glauber kinetics (model 02): the call is the attempt. Spin: agent i 's statement vectors in the regime-whitened 32-d basis. Field: the kickoff, direction $\hat{k}$; a is the excess alignment over decoy kickoffs (H125). Per agent: pre level P_i , settled level S_i (days 2–5), and a one-parameter rise on clock c (hours or own calls, from t_0 or from the read-out call). Rival R_L : the field arrives as a step at the read-out call (H08 gating) and nothing relaxes.

$$a_i(t) = P_i + A_i(1 - e^{-c_i(t)/\tau}), \quad T_{1/2} = \tau \ln 2$$

call clock: $\log T_{1/2,i}^H = \text{const} - \log r_i$ ($s = -1$); hour clock: $s = 0$

$$\Delta \text{SSE}_{\text{ro}} = \frac{\text{SSE}_H - \text{SSE}_C}{\text{SSE}_0 - \min(\text{SSE}_H, \text{SSE}_C)} \quad (\text{from the read-out call})$$



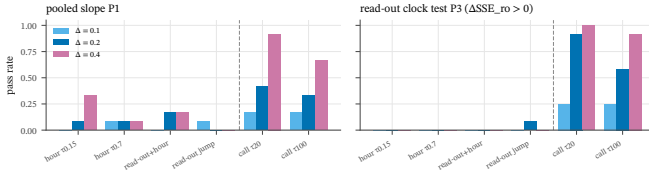
Observables. (a) Day-1 excess alignment of rising agents, normalized to each agent's late day-1 plateau (calls ≥ 64 after read-out, equal to 1 by construction), against own calls since the read-out call; both embedding models, 90% CI. Dashed: a call-clock rise with $\tau = 20$ calls. The data start at the top and decay. (b) The same against time since the kickoff, for the slower and faster halves of agents by call rate (bge).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Expect a new goal to show up in an agent's output at its first read-out call, about 75 s after the kickoff, not after a catch-up period. Speeding up an agent's calls will not make it adopt a goal faster; it is already adopted within one call. What does relax over the next hour is the kickoff's salience, from $2.4\times$ to $1\times$ the plateau. Do not judge adoption from the first statements, which overstate it.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 pooled slope near -1	+0.59 [-0.45, +1.75]; gte +0.42; power $\approx$ 0.35	inconclusive
P2 collapse CR < 1	0.65 / 0.19; ties at the floor	non-diagnostic
P3 $\Delta\text{SSE}_{\text{ro}} > 0$	-0.0009 [-0.0019, +0.0001]	not supported
P4 relaxation beyond read-out	immediate share 0.74 (relaxation worlds $\leq$ 0.20)	failed: R_L
S1 slope steeper in II-III	I +0.34, II-III +1.00	not supported
S4 rising share $\geq$ 0.5	0.60 / 0.57	supported
N1 G51 slope (21 agents)	all half times at the floor	failed
N2 NE38 prediction	Opus 5 $T_{1/2}$ 0.20 h; predictions at the floor	uninformative



Second figure. Synthetic swarms on the real skeleton (27 kickoffs, real statement, call and read-out times; 12 replicates per world and rise amplitude). The pooled slope reaches power 0.67–0.92 only at a rise of 0.4. The read-out-origin clock test separates hour and call clocks at a rise $\geq$ 0.2 and stays silent under a read-out jump.

GOAL PERIODS IN DETAIL

G04 d descriptive
G05 d descriptive
G06 d descriptive
G08 d descriptive
G10 $\times$ failed
G11 d descriptive
G12 $\times$ failed
G13 d descriptive
G16 d descriptive
G17 $\times$ failed

+18 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The first statements after a read-out likely restate the kickoff, so part of the spike is copying, not re-orientation. Half times below the 0.36-s grid floor are ties, which make per-kickoff slopes 0. The median rise (0.15–0.19) is below the amplitude where the slope test has power, so the slope miss is not a negative. Five kickoffs lack a previous non-holdout day. The rise curve is a post hoc description.

ROUND 2

Fit the clock of the spike's decay from the read-out call (it is large enough to test); measure resolved rises at single-agent role changes (NE38 and #51's later reassignments); run `confirm.py`.

- **H127-R1** The clock of the spike's decay.
- **H127-R2** Resolved rises.
- **H127-R3** Restatement.